MILITARY SPECIFICATION

LINES, RADIO FREQUENCY TRANSMISSION

(COAXIAL, AIR DIELECTRIC), GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 This specification covers the general requirements for radio frequency, coaxial, air dielectric copper and aluminum transmission lines (hereafter referred to as lines) (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FED STD NO. 356	-	Commercial Packaging of Supplies and Equipment.
WW-T-700/1	-	Tube, Aluminum, Drawn, 1100
PPP-B-640	-	Fiberboard, Corregated, Triple-wall.
PPP-T-66	-	Tape: Pressure-Sensitive Adhesive, Vinyl Plastic Film.
PPP-T-495	-	Tube, Mailing, and Filling.

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MIL-P-116	-	Preservation-Packaging, Methods of.
MIL-B-131	-	Barrier Materials, Watervaporproof, Flexible, Heat-Sealable.
MIL-L-3890/1	-	Lines, RF Transmission (Coaxial, Air Dielectric) 50 Ohms.
MIL-L-3890/2	-	Lines, RF Transmission (Coaxial, Air Dielectric) 75 Ohms.
MIL-C-5541	-	Chemical Films and Materials for Aluminum and Alum Alloys.
MIL-C-45662	-	Calibration System Requirements.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-1285	-	Marking of Electrical and Electronic Parts.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B188-55	-	Copper Bus Pipe and Tube, Seamless.
B210-67	-	Aluminum-Alloy Drawn Seamless Tubes.
B372-67	-	Seamless Copper and Copper-Alloy Rectangular Waveguide Tube.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Specification sheets. The individual requirements shall be as specified herein and in accordance with the applicable specification sheets.

3.2 First article. Lines furnished under this specification shall be products which have been tested and have passed the first article inspection specified in 4.5 (see 6.3).

3.3 Material. The material shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the line to meet the performance requirements of this specification. Acceptance or approval of any constituent shall not be construed as a guaranty of the acceptance of the finished product.

3.3.1 Aluminum. Aluminum shall conform to 1100, temper H18 in accordance with WW-T-700/1 and shall be chemically treated inside and outside with class 3 chemical treatment in accordance with MIL-C-5541.

3.3.1.1 Aluminum alloy. Aluminum alloy shall conform to the chemical composition and tensile property requirements for 6061 or 6063 of ASTM B210-67.

3.3.2 Copper. Copper shall conform to type OF or DLP of ASTM B372-67 or ASTM B188.

3.3.2.1 Copper alloy. Copper alloy shall conform to commercial bronze, 90 percent, of ASTM B372-67.

3.4 Design and construction. Lines shall be of the design, construction, and overall physical dimensions specified, (see 3.1).

3.4.1 Length. Unless otherwise specified (see 3.1), lines shall be furnished in 240-inch lengths. Lines ordered to specific or standard lengths shall conform to the tolerances in table I.

TABLE I. Length tolerances (applicable only to full length pieces).

Length	Tolerance, inch (all plus)
Standard (20 ft) - - - -	1/2
Specific	
Up to 20 ft- - - - - -	1/4
Over 20 ft - - - - - -	3/4

3.4.2 Bow. The natural bow of the lines shall not depart from a straight line more than 0.5 inch between any two points 10 feet apart on the external surface of the line. On lines less than 10 feet long, the allowable bow shall be .100 inch between two points 2 feet apart.

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3.4.3 Out-of-roundness. Unless otherwise specified (see 3.1), the outer and inner diameter of the conductors at any cross section shall not differ more than 1 percent from the specified diameter.

3.5 Attenuation. When lines are tested as specified in 4.7.2, the attenuation shall not exceed that specified (see 3.1).

3.6 Voltage standing wave ratio (VSWR). When lines are tested as specified in 4.7.3, the voltage standing wave ratio shall not exceed the value specified (see 3.1).

3.7 Characteristic impedance. When lines are tested as specified in 4.7.4, the characteristic impedance value shall not exceed the value specified (see 3.1).

3.8 Velocity of propagation. When lines are tested as specified in 4.7.5, the velocity of propagation value shall not be less than that specified (see 3.1).

3.9 RF power handling capability (when specified, see 3.1). When tested as specified in 4.7.6, the lines meet the following requirements after each test:

Attenuation - - - - - - - - - - - -	Shall be as specified (see 3.1).
Voltage standing wave ratio - - -	Shall be as specified (see 3.1).

3.10 Dielectric withstanding voltage. When tested as specifed in 4.7.7, the lines shall show no evidence of voltage breakdown.

3.11 Vibration, high frequency. When tested as specified in 4.7.8, the lines shall meet the following requirements:

Attenuation - - - - - - - - - - - -	Shall be as specified (see 3.1).
Voltage standing wave ratio- - -	Shall be as specified (see 3.1).

3.12 Pressurization (when specified, see 3.1). When lines are tested as specified in 4.7.9, there shall be no loss of pressurization as indicated by bubbles.

3.13 Marking. Marking shall be in accordance with MIL-STD-128S and shall be on each length of line at intervals of not more than 1.5 feet. The manufacturer's name or trademark may also be marked on the lines provided such letters are not expressly forbidden in the contract or order, and their height does not exceed 0.5 inch.

3.14 Workmanship. Lines shall be processed in such a manner as to be uniform in quality and shall be free from pits, corrosion, cracks, rough edges, chips, burrs, scratches, flaws and other defects that will affect life, serviceability, or appearance. The ends shall be cut square.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspection shall be established and maintained by the supplier. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-C-45662.

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4.2 Classification of inspections. The inspections specified herein are classified as follows:

(a) Materials inspection (see 4.4).
(b) First article inspection (see 4.5).
(c) Quality conformance inspection (see 4.6).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

4.4 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials listed in table II, used in fabricating the lines are in accordance with the applicable referenced specifications or requirements prior to such fabrication.

TABLE II. Materials inspection.

Material	Requirement paragraph	Applicable specification or publication
Aluminum	3.3.1	WW-T-700/1
Aluminum alloy	3.3.1.1	ASTMB210-67
Chemical treatment	3.3.1	MIL-C-5541
Copper	3.3.2	ASTM B372-67 or B188-55
Copper alloy	3.3.2.1	ASTM B372-67 or B188-55

4.5 First article inspection. First article inspection shall be performed by the supplier, after award of contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production. First article approval is valid only on the contract or purchase order under which it is granted unless extended by the Government to other contracts or purchase orders.

4.5.1 Sample size. Four lengths of a given size line shall be subjected to first article testing (6 1/8, 3 1/8, 1 5/8, 7/8, or 3/8 inch diameter lines.)

4.5.2 Inspection routine. Sample units shall be subjected to first article inspection specified in table III in the order shown. All sample units shall be subjected to the inspection of group I. The sample units shall then be divided equally into two groups of two sample units each and subjected to the inspection for their particular group.

4.5.3 Failures. One or more failures shall be cause for refusal to grant first article approval.

4.6 Quality conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspection.

4.6.1.1 Inspection lot. An inspection lot shall consist of all the lines of the same part number, produced under essentially the same conditions and offered for inspection at one time.

4.6.1.2 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table IV and shall be made on the same set of sample units, in the order shown.

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TABLE III. First article inspection.

Examination or test	Requirement paragraph	Method paragraph
Group I (all sample units)		
Visual and mechanical examination	3.1,3.3 thru 3.4 incl. 3.13 and 3.14	4.7.1
Attenuation	3.5	4.7.2
VSWR	3.6	4.7.3
Characteristic impedance	3.7	4.7.4
Velocity of propagation	3.8	4.7.5
Group II (two sample units)		
RF power handling capability (when specified)	3.9	4.7.6
Dielectric withstanding voltage	3.10	4.7.7
Group III (two sample units)		
Vibration, high frequency	3.11	4.7.8
Pressurization (when specified)	3.12	4.7.9

TABLE IV. Group A inspection.

Examination or test	Requirement paragraph	Method paragraph	AQL percent defective	
			Major	Minor
Visual and mechanical examination	3.1,3.3 thru 3.4 incl 3.13 and 3.14	4.7.1	1.0	4.0
Attenuation	3.5	4.7.2	1.0	
VSWR	3.6	4.7.3	1.0	
Characteristic impedance	3.7	4.7.4	1.0	
Velocity of propagation	3.8	4.7.5	1.0	

4.6.1.2.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality levels (AQL) shall be as specified in table IV. Major and minor defects shall be as defined in MIL-STD-105.

4.6.1.2.2 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.6.1.3 Group B inspection. Group B inspection shall consist of the test specified in table V and shall be made on sample units which have been subjected to and have passed the group A inspection.

TABLE V. Group B inspection.

Test	Requirement paragraph	Method paragraph
Dielectric withstanding voltage	3.10	4.7.7

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4.6.1.3.1 Sampling plan. The sampling plan shall be in accordance with MIL-STD-105 for special inspection level S-4. The sample size shall be based on the inspection lot size from which the sample was selected for group A inspection. The AQL shall be 6.5 percent defective.

4.6.1.3.2 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.6.1.3.3 Disposition of sample units. Sample units which have passed all the group B inspection may be delivered on the contract or purchase order, if the lot is accepted.

4.6.2 Inspection of preparation for delivery. Except when commercial packaging is specified, the sampling and inspection of the preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129. The inspection of commercial packaging shall be as specified in the contract or purchase order (see 6.2).

4.7 Method of examination and test.

4.7.1 Visual and mechanical examination (see 3.1, 3.3 through 3.4 incl., 3.13 and 3.14). Lines shall be examined to verify that the material, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements.

4.7.2 Attenuation (see 3.5). The attenuation of the lines shall be measured over the specified frequency range (see 3.1). The test equipment(s) shall be capable of providing a continuous measurement of attenuation over the required frequency ranges. A means shall be provided for producing a permanent record of the line attenuation versus frequency. The length of the test sample shall be 10 feet (minimum).

4.7.3 VSWR (see 3.6). The VSWR of the lines shall be measured over the specified frequency range (see 3.1). The test equipment(s) shall be capable of providing a continuous measurement of VSWR over the required frequency ranges. A means shall be provided for producing a permanent record of the line VSWR versus frequency. If VSWR is not directly measured; that is, if return loss is measured and VSWR is calculated from that measurement, the permanent record shall indicate the worst case VSWR numerically for each frequency band and shall provide the calculation used to obtain the calculated VSWR. The length of the test sample shall be 100 feet (minimum).

4.7.4 Characteristic impedance (see 3.7). The characteristic impedance of the lines shall be measured with Time Domain Reflectometry (TDR) equipment or other methods that are approved by the procuring activity. The TDR equipment shall have a total system response of 150 pico-seconds and shall provide for a resolution of 0.1 ohm throughout the length of the line. The test results shall be displayed on calibrated rectangular coordinates and a permanent record of the test results shall be produced. A section of air line with a calibrated impedance of 50.0 ±0.15 ohms is required as a reference.

4.7.5 Velocity of propagation (see 3.8). The velocity of propagation of the lines shall be determined as follows:

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4.7.5.1 Test equipment. The following test equipment shall be utilized:

(a) A microwave sweep oscillator with a frequency range that covers the specified frequency range of the assembly.
(b) An analyzer with suitable readout features.
(c) A phase bridge with a measurement accuracy of ±0.1 degree.
(d) A calibrated section of air line whose length equals the length of the line under test.

4.7.5.2 Test procedures:

(a) Calibrate the system over that portion of the specified frequency range of the line that will be within the limits of 4.7.5.1(c) with required adapters inserted in the leg of the phase bridge to be used for the line.
(b) Insert the line into the appropriate leg of the phase bridge and an equal length of air line in the other leg. Note the length (L) of the line.
(c) Using the phase bridge, adjust the electrical length of the air line leg to obtain a phase balance across the frequency range of 4.7.5.2(a). Note the added length (ΔL).
(d) Calculate velocity of propagation (VP) of calibration line as follows:

$$VP = \frac{L}{L + \Delta L} \times 100\%$$

(e) Sweep over the specified frequency of the assembly. Note maximum deviation (Ø in degrees) from calibration line.
(f) Calculate minimum velocity of propagation VP (min) as follows:

$$VP\ (Min) = \frac{L}{L + \Delta L + \frac{\emptyset C}{360\ f}}$$

where:
C is the velocity of propagation in free space.
f is the frequency (in Hz) at which Ø was measured.

4.7.6 RF power handling capability (see 3.9)(when specified, see 3.1). The power handling capability shall be determined by subjecting the lines to average RF power at the specified lowest frequency, VSWR of unity (as nearly as possible), one atmosphere of dry air (40% relative humidity maximum) pressure at sea level and the highest temperature continuously for 30 minutes or until the line temperature stabilizes, whichever comes first. The lines shall also be subjected to peak power at the specified lowest frequency and the highest temperature for 30 minutes or until the line temperature stabilizes, whichever comes first. The lines shall be terminated in a match load. Following each power test, attenuation and VSWR shall be measured as specified in 4.7.2 and 4.7.3, respectively.

4.7.7 Dielectric withstanding voltage (see 3.10). The lines shall be tested in accordance with method 301 of MIL-STD-202. The following details shall apply:

(a) Conditions of test. One atmosphere of dry air (40% relative humidity) pressure at sea level.
(b) Duration of application. When the specified test voltage (see 3.1) is:
  (1) Less than 1,500 volts rms, the voltage shall be instantaneously applied.
  (2) Above 1,500 volts rms, the voltage shall be applied at approximately 500 volts per second until the rated test voltage is reached or breakdown occurs.
(c) Points of application. The 60-cycle voltage shall be applied between the inner conductor and the outer conductor of the line.

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4.7.8 Vibration, high frequency (see 3.11). The lines shall be tested in accordance with method 204 of MIL-STD-202. The following details shall apply:

(a) Method of mounting - Clamp the lines to the table. No part of the lines shall touch any object other than the clamps (one clamp at each end and one in the middle if required).
(b) Test condition letter - B.
(c) Examination after test - Attenuation and VSWR shall be measured as specified in 4.7.2 and 4.7.3.

4.7.9 Pressurization (when specified) (see 3.12). With the line ends properly sealed, the line shall be subjected to the internal gas pressure specified (see 3,1) for at least 20 seconds while immersed in tap water at a temperature of approximately 20°C. Unless otherwise specified (see 3.1), the internal gas pressure shall be 30 ±1 pound per square inch. During immersion, the lines shall be observed for evidence of leakage as indicated by escaping air bubbles.

5. PREPARATION FOR DELIVERY

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, or as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Transmission lines shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Transmission lines shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative application. Preservatives shall not be used.

5.1.1.4 Unit packaging. Transmission lines shall be unit packaged one each in accordance with method IA of MIL-P-116 as follows: The open ends of each transmission line shall be covered by a water-vaporproof barrier material conforming to MIL-B-131, class 1. This barrier material shall extend a minimum of 6 inches back from each open end and shall be secured and sealed with tape conforming to PPP-T-66, type I. When the opening exceeds the width of the tape, cross strips shall be used to reinforce the barrier material. The unit container shall conform to the level of packing specified (see 5.2).

5.1.1.5 Intermediate packaging. Not required.

5.1.2 Level C. Transmission lines shall be clean, dry, and packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity. This level may conform to the supplier's commercial practice when such meets the requirements of this level. Unless otherwise specified in the contract or purchase order (see 6.2), the quantity per unit package shall be one each.

5.2 Packing. Packing shall be level A, B, or C or as specified (see 6.2).

5.2.1 Level A. Each transmission line shall be packed in a waterproof tube conforming to PPP-T-495, type II except that the minimum wall thickness shall be 1/2-inch collar blocks containing 1/4-inch oversize holes and spaced no more than 2 feet apart shall be used to position each transmission line within the container. If the container requirements are exceeded weather resistant fiberboard boxes conforming to PPP-B-640, class 2 shall be used. Closure, sealing, and reinforcement requirements shall be in accordance with the appendix of this box specification.

5.2.2 Level B. This packing shall be as specified in 5.2.1 except that the tubes shall conform to PPP-T-495, type I and the boxes shall conform to PPP-B-640, class 1.

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5.2.3 Level C. The transmission lines shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirements of this level.

5.2.4 Unitized loads. Shipments of two or more packed transmission lines to a single destination shall be unitized by securely banding or strapping the packs together or by farther packing in a suitable consolidation container. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.3 Marking. In addition to any special marking required by the contract or purchase order (see 6.2), each container shall be marked in accordance with MIL-STD-129.

5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2, and 5.2.3) shall be of a minimum tare and cube consistent with the protection required and contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.6.2.

5.4.3 Army procurements. For Army procurements commercial packaging (including unit packaging, packing, and marking) shall be in accordance with Federal Standard No. 356.

6. NOTES

6.1 Intended use. Transmission lines are used to transmit RF energy in the microwave region of the spectrum.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Title, number, and date of the applicable specification sheet and the military part number (see 3.1).
(c) Inspection of commercial packaging (see 4.6.2).
(d) Levels of preservation-packaging and packing required (see 5.1 and 5.2).
(e) Quantity per unit package, if other than one each (see 5.1.2).
(f) Special marking, if required (see 5.3).

6.3 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MU, MI, SG, ME
Navy - OS
Air Force - 11, 17
DSA - ES

User activities:
Army - AT, WC
Navy - MC, SH, AS
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

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